

17 Modelling

7 primary questions | Oldest to newest

2026 objectives: 4.1

14 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

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CA1-P2 April 2006 Q1	5 marks
CA1-P2 April 2009 Q3	8 marks
CA1-P2 September 2009 Q7	27 marks
CA1-P2 September 2014 Q3	12 marks
CA1-P2 April 2015 Q3	9 marks
CA1-P2 September 2016 Q7	25 marks
CA1-P2 September 2018 spare Q6	19 marks

CA1-P2 | April 2006 | Q1

[Current topic](#) | [5 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 1** Outline the reasons why a general insurance company models its claims experience. [5]

CA1-P2 | April 2009 | Q3

[Current topic](#) | [8 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 3** An actuary has been asked to carry out a small, one-off investigation for a client, which will require the use of a model.
- Discuss the factors the actuary should take into account when considering how to approach the development of this model and choosing the source of the model. [8]

CA1-P2 | September 2009 | Q7

Current topic | 27 marks total | Original PDF p4

Related: Ch 25, Ch 30.

- 7** A company is considering purchasing a large office building located in a large city and is assessing the options available to manage the risks associated with the purchase.
- | (i) Give six examples of risks that could lead to the loss of use of the entire building. [3]
- (ii) Discuss the management of these risks. [4]
- (iii) Discuss the requirements of a model to be used to assess these risks. [9]
- (iv) Outline the steps required to evaluate these risks using a model. [5]
- (v) Discuss the advantages and disadvantages of using the following techniques to create a model:
- (a) Stochastic modelling
- (b) Stress testing
- (c) Scenario analysis
- [6]
- [Total 27]

CA1-P2 | September 2014 | Q3

Current topic | 12 marks total | Original PDF p2

Related: Ch 32.

- 3** A small general insurance company currently sells only one line of business. The company's strategic plan is to grow rapidly by acquiring other insurance companies.

The company's owners have identified a target company, and now wish to determine an offer price for the target.

- (i) Outline the requirements of an actuarial model to place a value on the target company. [7]
- (ii) Discuss why the owners may offer a different price to that suggested by the model. [3]
- (iii) Discuss why the company may be able to operate more efficiently after completing the acquisition. [2]

[Total 12]

CA1-P2 | April 2015 | Q3

Current topic | 9 marks total | Original PDF p2

Related: Ch 1.

- 3** A large bank is about to launch a new product designed for sophisticated investors. The product will involve direct investment in a number of markets, together with associated derivatives.

The bank also intends to offer an insurance policy linked to the new product so that investors who take out insurance will receive protection against large falls in the value of their holding in the new product.

When setting the premium rates for the insurance policy, the bank has used advanced stochastic techniques to model a range of financial outcomes. This model is only accessible to and understood by a small specialist team within the bank and is therefore treated as a “black box” by all others in the bank.

The results of the model suggest that the bank will be able to invest the insurance premiums in such a way that the invested premiums will be almost certain to meet any claims. The Chairman of the bank interprets this to mean that this insurance policy can, in effect, achieve guaranteed profits.

Discuss why it may not be appropriate for the Chairman to accept the results of the model in this way.

[9]

CA1-P2 | September 2016 | Q7

Current topic | 25 marks total | Original PDF p4

Related: Ch 1, Ch 31.

- 7** A life insurance company is performing a supervisory valuation of its many lines of business.

Most material lines of business are valued using an industry-standard actuarial modelling system, called ELL, purchased from an external actuarial consultancy. Some material lines of business are, however, still valued using spreadsheets.

ELL is the company's strategic preferred system for all actuarial calculations across its operations. It is run and controlled by the company's central actuarial team, and receives data feeds directly from the company's administration systems.

Spreadsheets are controlled by the valuation team, separate from the central actuarial team, and rely on data supplied manually by the relevant administration teams.

- (i) Discuss the advantages and disadvantages to the company of performing its supervisory valuation using ELL compared with the spreadsheets. [13]

One of the material lines of business which is still valued on a spreadsheet is required to be valued on ELL instead, in time for the next supervisory valuation. Relevant project conversion teams will be put in place to plan and implement this conversion. Implementation can occur only if the valuation team's actuary provides written approval that the revised system is fit for the purpose of the next supervisory valuation. The valuation team's actuary is not part of the project conversion teams.

- (ii) Discuss the steps and investigations that the valuation team's actuary should perform before providing written approval of the revised system. [8]

At the first supervisory valuation to be published following conversion of this line of business, the valuation team's actuary is unable to reconcile the ELL reserves for this line of business with the reserves that would have been obtained using the spreadsheet. The Directors are about to need to sign off the published valuation to meet the deadline set by the regulator.

- (iii) Outline how the valuation team's actuary should now proceed. [4]
[Total 25]

CA1-P2 | September 2018 spare | Q6

Current topic | 19 marks total | Original PDF p3

2018 spare paper (X18), distinct from the September main sitting. Original paper and matching spare examiner report restored from the supplied archive. Related: Ch 19, Ch 36.

- 6** (i) Set out the steps involved in stochastic modelling. [7]

An insurance company is developing a stochastic economic capital model. The model will be used to measure the risk in a portfolio of corporate bonds. The risk will be modelled by simulating credit spreads (the difference in yield between bonds of different credit ratings) and credit rating transitions (the move from one rating to another of a specific bond). The insurer has obtained the following data to use for the calibration of the model:

- Credit spreads: Money weighted credit spreads for corporate bond indices since 2000 split by credit rating.
- Credit rating transitions: Issuer weighted credit rating transitions since 1970 split by credit rating.

- (ii) Explain why there is a correlation between credit spreads and credit transitions. [4]

- (iii) Discuss the issues in using this data in setting those correlations. [6]

- (iv) Discuss alternative approaches to set the correlation between credit spreads and credit transitions. [2]

[Total 19]